GEOFCAN 2025 AVIGNON

pyGIMLi

Geophysical Inversion & Modelling Library

GEOFCAN
Géophysique des Sols
et
des Formations Superficielles

Short course on Petrophysical Joint Inversion

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2025-06-04

Agenda

- **Welcome Introduction**
- **Workshop presentation**
- Short introduction to pyGIMLi
- Presentation of 4PM PJI
- History of the 4PM PJI
- **Get started with individual inversions**
- Data format
- Connection to Renku
- Introduction to the different folders and files
- Make mesh and individual inversions

☕ COFFEE BREAK

- **Petrophysical joint inversion – First run with a field data set**
- Which parameter to use/modify/test? (start porosity, poro_min, poro_max, regularization parameters, etc.)
- Which equation to use? how?
- How to interpret your 3 or 4 phases estimations?
- **Petrophysical joint inversion**
- Try out the PJI with your own data or different data provided
- **Go further synthetic dataset**

introducing pyGIMLi

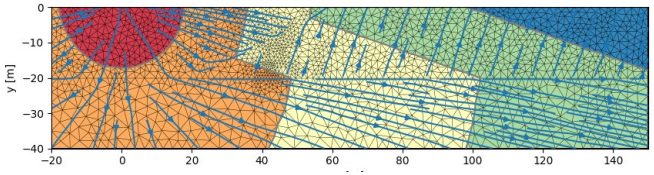
is rather a box full of tools and not a specific tool

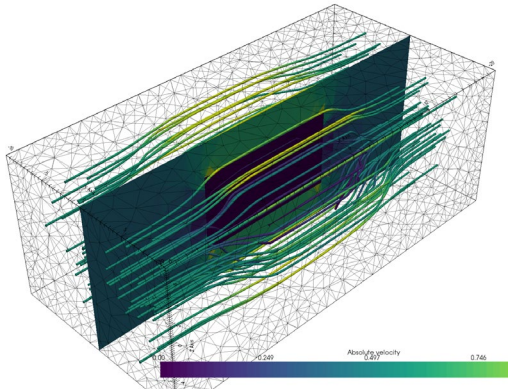
aims at making:

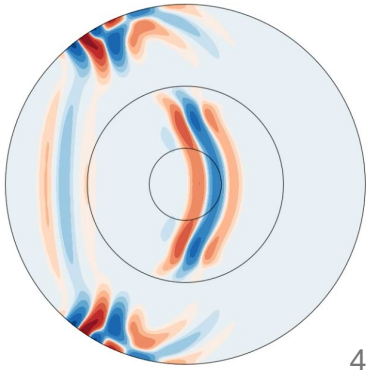
- easy things easy
- hard things possible
- everything transparent and reproducible

pyGIMLi is a versatile open-source toolbox with:

- management tools for **structured and unstructured meshes** in2D & 3D
- computationally efficient **finite-element and finite-volume solvers**
- **various geophysical forward operators**: ERT/IP, Traveltime, Gravimetry, Magnetics, SP, EM
- frameworks for **constrained, joint and process-based inversions** with **region-specific regularization**
- **open-source, platform compatible**, documented & tested code
- suitability for **teaching & reproducible research**
- v1.0 published in *Computers and Geosciences* () (among *Most Downloaded* papers, > 430 citations, > 120 uses in peer-reviewed papers)

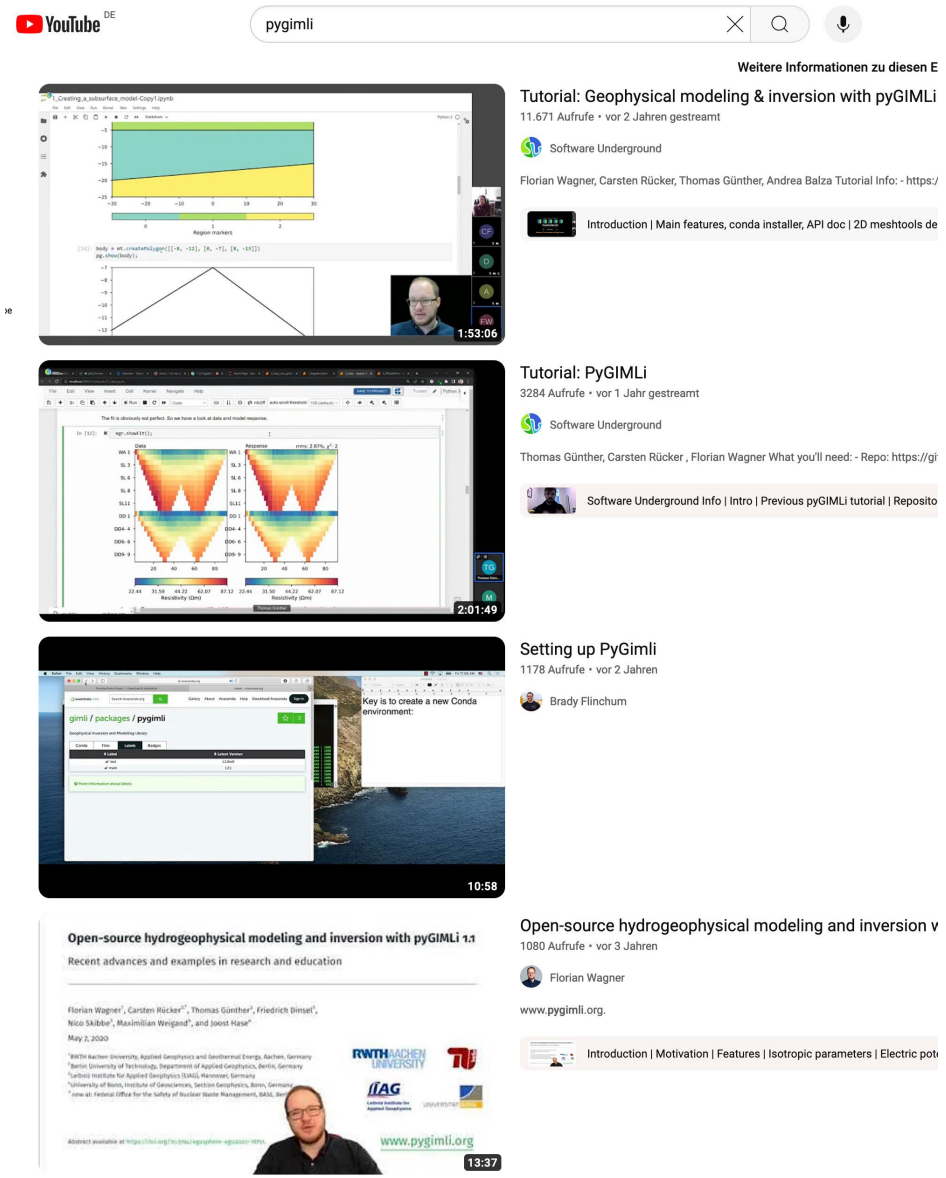






Existing tutorials

- creating **geometries & meshes**, **modeling** PDEs, **synthetic data**, **inversion** (also with **external forward operator**).
- Transform 2021: fundamental `pyGIMLi` objects (`Mesh`, `DataContainer`, matrix types, etc.), **geostatistical** vs. smoothness regularization, treatment of subsurface **regions**, adding **prior data**.
- Transform 2022 : invert **real-life 3D data** ( ) to tweak your **inversion beyond the standard** practice. SEG webinar 2024 Hübner et al., 2017
- Gelmon 2025 : **ERT monitoring** data processing & inversion.



About pyGIMLi

pyGIMLi is an open-source library for modelling and inversion and in geophysics. The object-oriented library provides management for structured and unstructured meshes in 2D and 3D, finite-element and finite-volume solvers, various geophysical forward operators, as well as Gauss-Newton based frameworks for constrained, joint and fully-coupled inversions with flexible regularization.

What is pyGIMLi suited for?

- analyze, visualize and invert geophysical data in a reproducible manner
- forward modelling of (geo)physical problems on complex 2D and 3D geometries
- inversion with flexible controls on a-priori information and regularization
- combination of different methods in constrained, joint and fully-coupled inversions
- teaching applied geophysics (e.g. in combination with Jupyter notebooks)

What is pyGIMLi **NOT** suited for?

- for people that expect a ready-made GUI for interpreting their data

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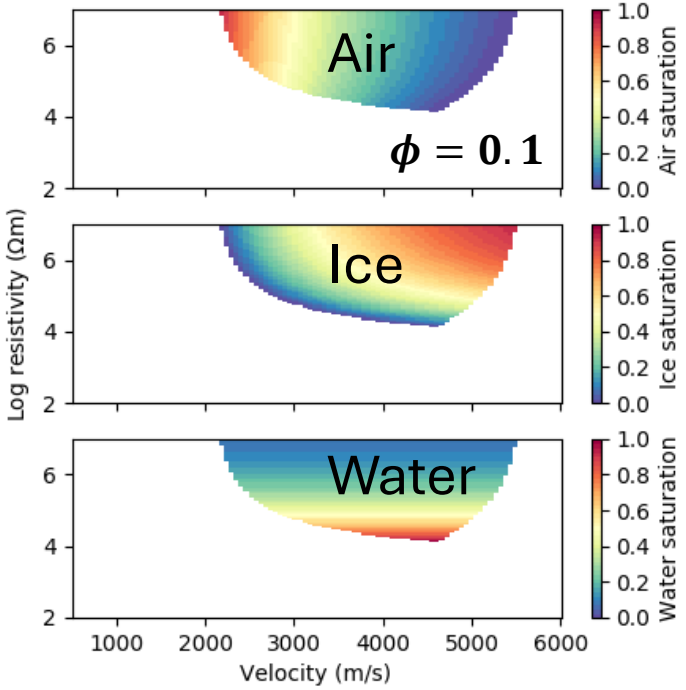
What is the 4-phase model by Hauck et al. 2011?

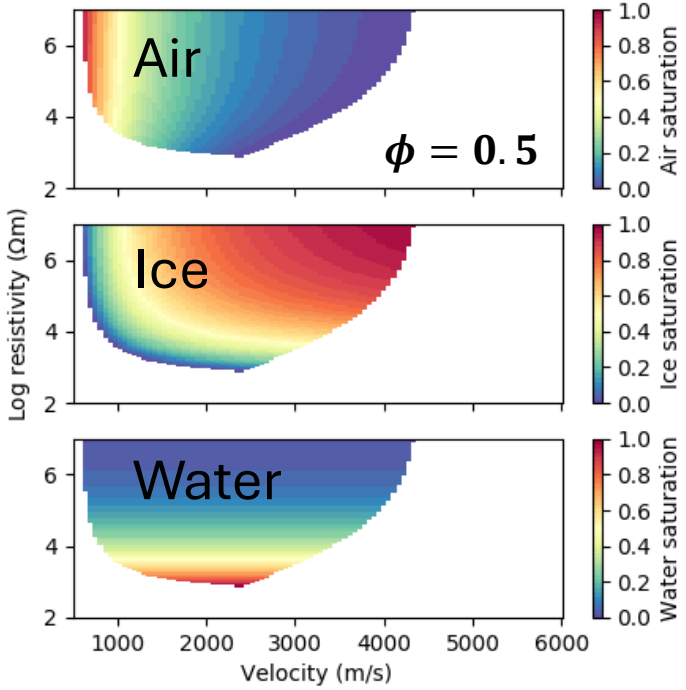
The **4-phase model** (4PM) by Hauck et al. (2011) is based on: **petrophysical relationships** linking ice, water and air fractions (f_i, f_w, f_a) with the resistivity $\boldsymbol{\rho}$, P-wave velocity $\boldsymbol{v}$ and porosity $\boldsymbol{\phi}$

Timur/Wyllie $\frac{1}{\boldsymbol{v}} = \frac{f_r}{v_r} + \frac{f_w}{v_w} + \frac{f_a}{v_a} + \frac{f_i}{v_i}$

Archie $\boldsymbol{\rho} = a\,\rho_w\,\phi^{-m}\left(\frac{f_w}{\phi}\right)^{-n}$

Volume conservation $f_r + f_i + f_w + f_a = 1$





→ White areas indicate no solution

Petrophysical joint inversion (PJI)

https://github.com/florian-wagner/four-phase-inversion

- Wagner et al. (2019)

based on the 4-phase model (4PM) by Hauck et al. 2011

Geophysical Journal International

Geophys. J. Int. (2019) **219**, 1866–1875 doi: 10.109
Advance Access publication 2019 September 03
GJI Marine Geosciences and Applied Geophysics

Quantitative imaging of water, ice and air in permafrost systems through petrophysical joint inversion of seismic refraction and electrical resistivity data

F.M. Wagner[1], C. Mollaret,[2] T. Günther,[3] A. Kemna[1] and C. Hauck[2]

[1]*Institute of Geosciences, Geophysics Section, University of Bonn, Bonn, Germany. E-mail: mail@fwagner.info*
[2]*Department of Geosciences, University of Fribourg, Fribourg, Switzerland*
[3]*Leibniz Institute for Applied Geophysics, Hannover, Germany*

Accepted 2019 August 31. Received 2019 June 17; in original form 2019 August 13

github.com/florian-wagner/four-phase-inversion

README License

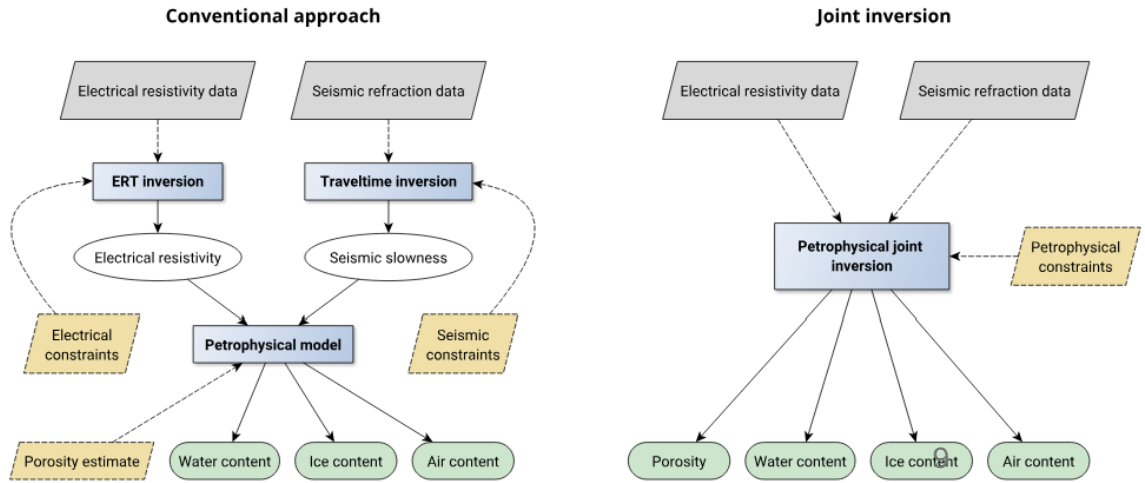
Imaging of water, ice, and air in permafrost systems through petrophysical joint inversion of seismic refraction and electrical resistivity data

by Florian Wagner, Coline Mollaret, Thomas Günther, Andreas Kemna, and Christian Hauck

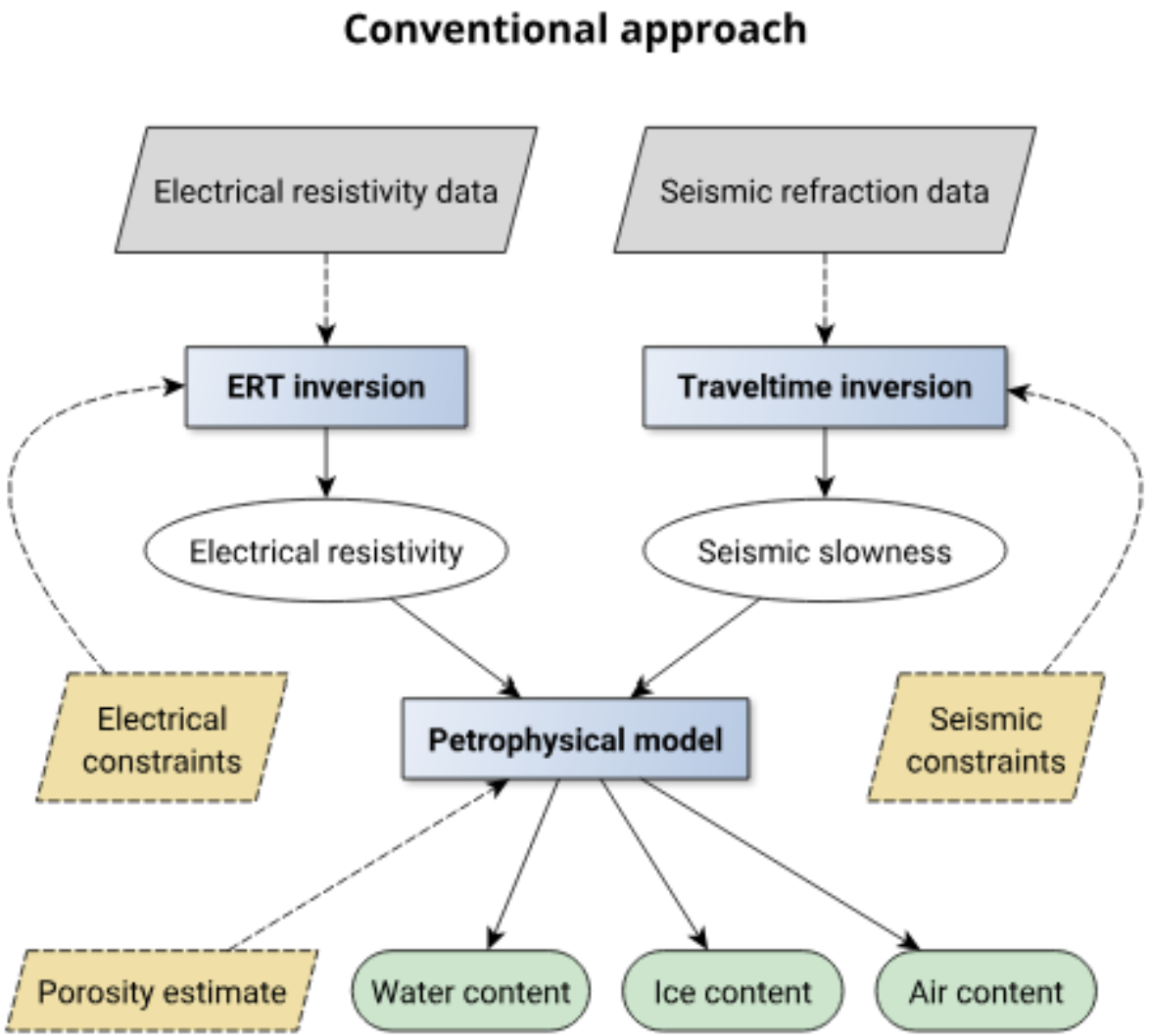
DOI 10.1093/gji/ggz402 license BSD powered by pyGIMLi

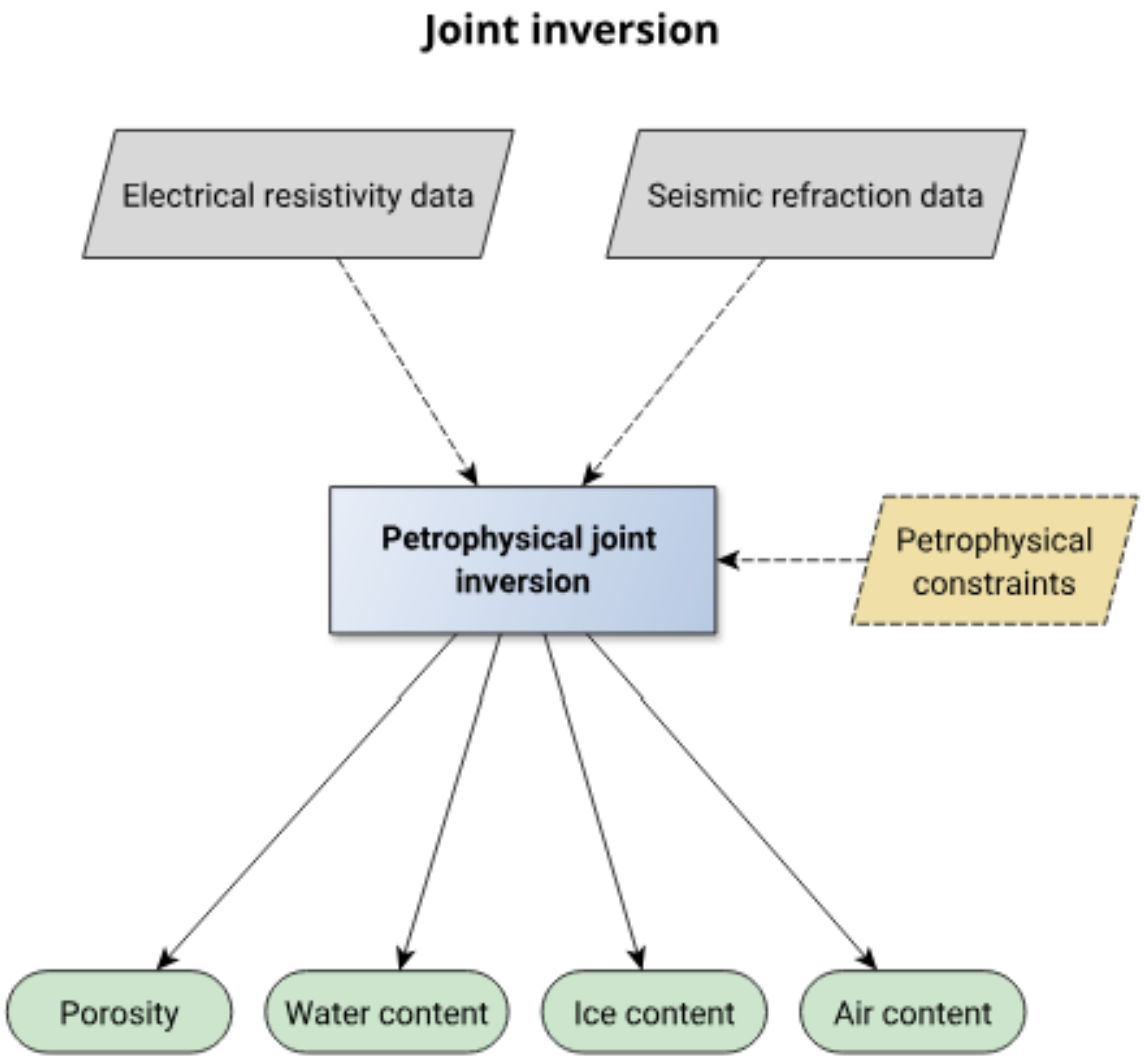
This repository contains the data and code to reproduce all results and figures published in the following paper:

> Wagner, F. M., Mollaret, C., Günther, T., Kemna, A., & Hauck, C. (2019). Quantitative imaging of water, ice, and air in permafrost systems through petrophysical joint inversion of seismic refraction and electrical resistivity data. Geophysical Journal International, 219, 1866-1875, doi:10.1093/gji/ggz402.



Petrophysical joint inversion (PJI)





Motivation

- **Original goal: quantification of the ice content in permafrost environment**
- Standard borehole temperature monitoring gives little insight into the ice content
- **Quantitative estimation of the ice content is rare**
- Spatial heterogeneity is challenging to retrieve (of porosity, ice content, etc.)
- Estimation of ice content can serve as calibration or validation of modelling studies





Peer-reviewed papers using PJI

AR: Archie
SC: Surface conduction factor
GM: Geometric mean model

- Wagner et al. (2019) based on the 4-phase model (4PM) by Hauck et al. 2011: PJI-AR
- Mollaret et al. (2020) compared **4 resistivity petrophysical equations**: PJI-AR, 2 PJI-AR+SC, PJI-GM

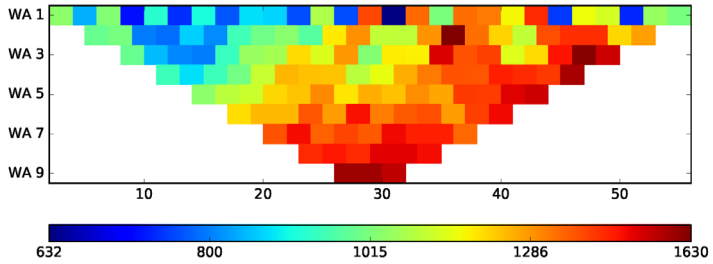
Many further applications published (not exhaustive list):

De Pasquale et al. (2020)	Rock glacier and relict (Andes, CL)	PJI-AR
Steiner et al. (2021)	Bedrock (Sonnblick, AT)	PJI-AR constrained by GPR
Steiner et al. (2022)	Landfills (AT)	**3PM** PJI-WS
Carrier et al. (2022)	Limestone Pillars (FR)	**3PM** PJI-AR
Pavoni et al. (2023)	2 rock glaciers (CH)	PJI-AR + structural coupling
Morard et al. (2024)	Stockhorn plateau (CH)	PJI-AR (monitoring with fixed rock content)
Maierhofer et al. (2024)	Cervinia plateau (IT)	PJI-AR+SC (surface conduction)
Wee et al. (2024)	Rock glacier (Gruben, CH)	PJI-GM
Mathys et al. (2025)	21 profiles (various landforms, KG, TJ)	PJI-AR & PJI-GM

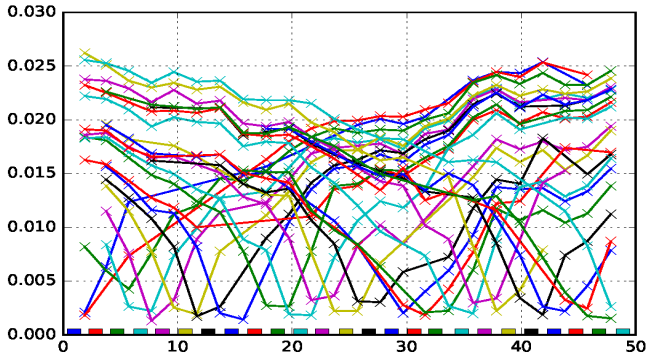
- **Several conference abstracts**: e.g. Klahold et al. 2021, Hilbich et al. 2025 Time-lapse PJI

Petrophysical joint inversion

Apparent resistivity data



Seismic travel time data



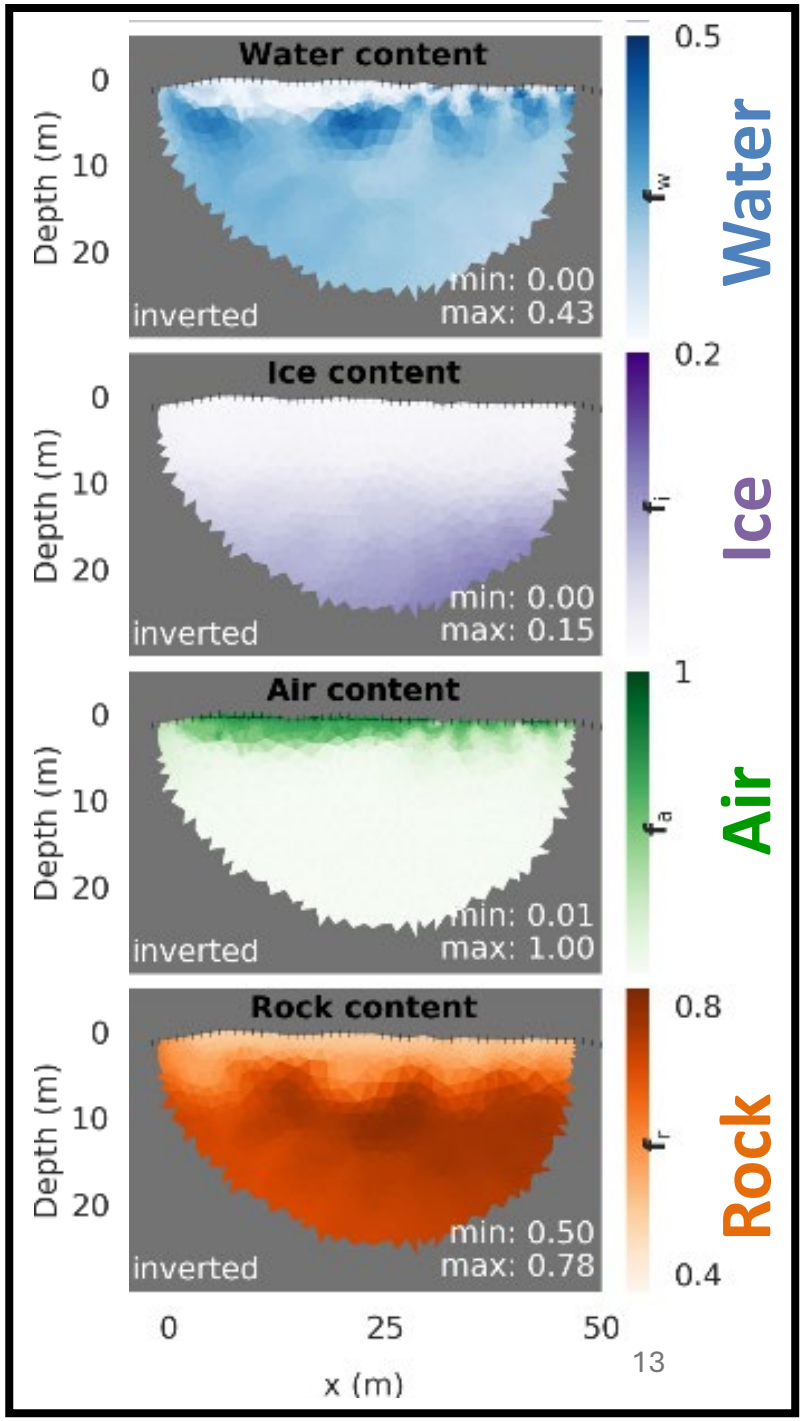
Petrophysical equations

$$\rho = \rho_w (1 - fr)^{-m} \left(\frac{f_w}{1 - fr} \right)^{-n}$$

$$\frac{1}{v} = \frac{f_r}{v_r} + \frac{f_i}{v_i} + \frac{f_w}{v_w} + \frac{f_a}{v_a}$$

Petrophysical Joint Inversion





Petrophysical joint inversion

Volume conservation

$$f_r + f_i + f_w + f_a = 1$$

Link to the P-wave velocity

Timur/Wyllie

$$\frac{1}{v} = \frac{f_r}{v_r} + \frac{f_i}{v_i} + \frac{f_w}{v_w} + \frac{f_a}{v_a}$$

Link to electrical resistivity

Archie

$$\rho = \rho_w (1 - f_r)^{-m} \left(\frac{f_w}{1 - f_r} \right)^{-n}$$

Geometric mean model

$$\rho = {\rho_r}^{f_r} . {\rho_i}^{f_i} . {\rho_w}^{f_w} . {\rho_a}^{f_a}$$

Input data: $\boldsymbol{d} = [t, \log(\rho_a)]^T$

Parameter data: $\boldsymbol{m} = [f_i, fw, fa, fr]^T$

The **Jacobian/Sensitivity matrix** J links d and m: $d = J\, m$

$$J = \begin{bmatrix} \frac{\partial t}{\partial f_w} & \frac{\partial t}{\partial fi} & \frac{\partial t}{\partial fa} & \frac{\partial t}{\partial fr} \\ \frac{\partial \log(\rho_a)}{\partial fw} & \frac{\partial \log(\rho_a)}{\partial fi} & \frac{\partial \log(\rho_a)}{\partial fa} & \frac{\partial \log(\rho_a)}{\partial fr} \end{bmatrix} = \begin{bmatrix} \frac{\partial t}{\partial s}\frac{\partial s}{\partial f_w} & \frac{\partial t}{\partial s}\frac{\partial t}{\partial fi} & \frac{\partial t}{\partial s}\frac{\partial t}{\partial fa} & \frac{\partial t}{\partial s}\frac{\partial t}{\partial fr} \\ \frac{\partial \log(\rho_a)}{\partial \rho}\frac{\partial \rho}{\partial fw} & \frac{\partial \log(\rho_a)}{\partial \rho}\frac{\partial \rho}{\partial fi} & \frac{\partial \log(\rho_a)}{\partial \rho}\frac{\partial \rho}{\partial fa} & \frac{\partial \log(\rho_a)}{\partial \rho}\frac{\partial \rho}{\partial fr} \end{bmatrix}$$

PJI: minimisation of the objective function

Input data: $\boldsymbol{d} = [t, \log(\rho_a)]^{\mathsf{T}}$
Parameter data: $\boldsymbol{m} = [f_i, fw, fa, fr]^{\mathsf{T}}$

The **Jacobian/Sensitivity matrix** J links d and m: $d = J\,m$

Objective function:

$$\|\boldsymbol{W}_{\mathrm{d}}(\boldsymbol{d} - \boldsymbol{\mathcal{F}}(\boldsymbol{m}))\|_2^2 + \alpha^2\|\boldsymbol{W}_{\mathrm{m}}\boldsymbol{m}\|_2^2 + \beta^2\|\boldsymbol{W}_{\mathrm{p}}^{\mathrm{sum}}\boldsymbol{p} - \mathbf{1}\|_2^2 \rightarrow \min.$$

Misfit between observed data and the model response

Smoothness regularization

Volume conservation

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It’s almost your turn to get started

- 1_make_mesh.ipynb
- 2_ERT+RST_inversions.ipynb
- 3_petrophysical_joint_inversion.ipynb

Input data format

http://resistivity.net/bert/data_format.html



Boundless electrical resistivity tomography

Navigation

Installation
The unified data format
Apps
Examples
API documentation
Source code

Quick search

Go

The unified data format

All projects in our working group use a unified format such that data are transpararent and transportable. There are a lot of import and export filters for third party software or hardware, but internally the following structure is used. Since the most applications use multi-electrode systems they are based on the way they are addressed (and computed in modeling).

Every data set consists of two mandatory parts: electrode definitions and arrangement definition. Subsequently the number of electrodes, their positions (x z for 2D, x y z for 3D), the number of data and their definitions are given. Comments may be places after the # character.

After the number of data the definition for each datum is given row-wise by the electrode numbers for the current electrodes A and B (C1 and C2) and the potential electrodes M and N (P1 and P2). Please not all electrode number starting from one. Electrodes with a zero number will be treated as infinity electrodes. They may be followed by other attributes (see list below). Standard is the apparent resistivity and optional the relative error as exemplified for a tiny dipole-dipole survey.

```
6# Number of electrodes
# x z position for each electrode
0     0
1     0
2     0 # loose ground
3     0
4     0
5     0
6# Number of data
# a b m n rhoa
1   2   3   4  231.2
2   3   4   5  256.7
3   4   5   6  312.8
```

Data available

rhoa: apparent resistivities
tt: travel times

/ data /

Name
CH_COR_V01_2007-07-13_01_rhoa.txt
CH_COR_V01_2007-07-13_01_tt.txt
CH_DOL_V01_2016-08-11_01_rhoa.txt
CH_DOL_V01_2016-08-11_01_tt.txt
CH_LAP_MH1_2010-08-24_01_rhoa.txt
CH_LAP_MH1_2010-08-24_01_tt.txt
KG_ABR_P03_2021-08-22_01_rhoa.txt
KG_ABR_P03_2021-08-22_01_tt.txt

Mollaret et al. 2020 (Frontiers)

Mathys et al. 2025 (The Cryosphere)

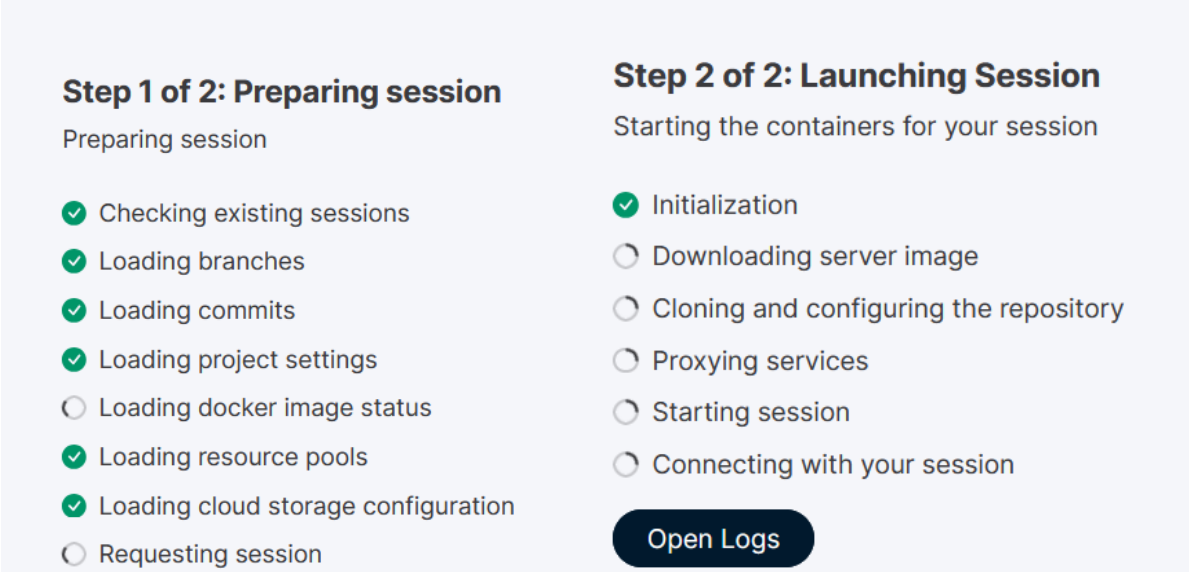
How to connect to the Renku platform?

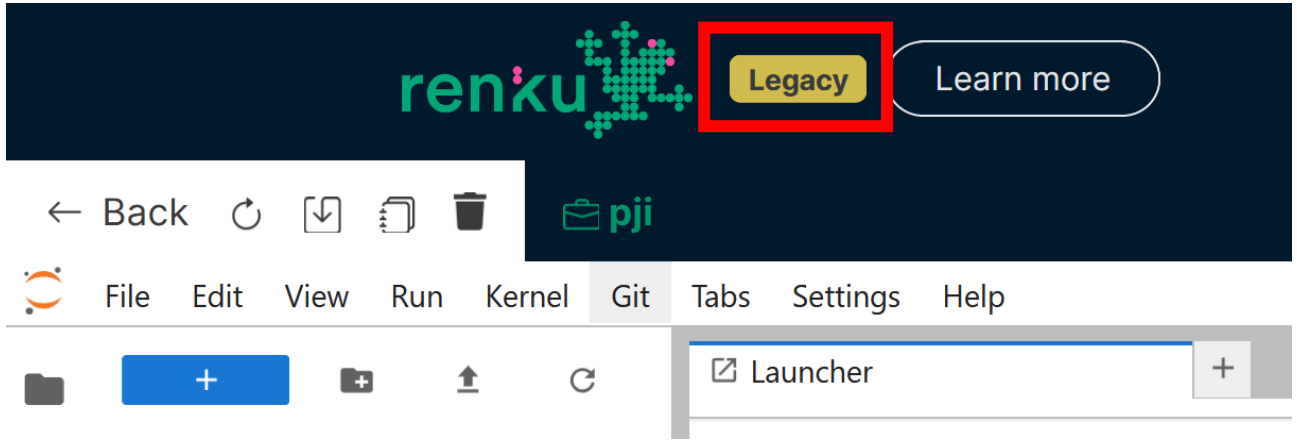
- Renku is a collaboratory platform: **https://renkulab.io**
- The IT resources for today's workshop are gently provided through the IT service of the University of Fribourg.

- Link to connect:

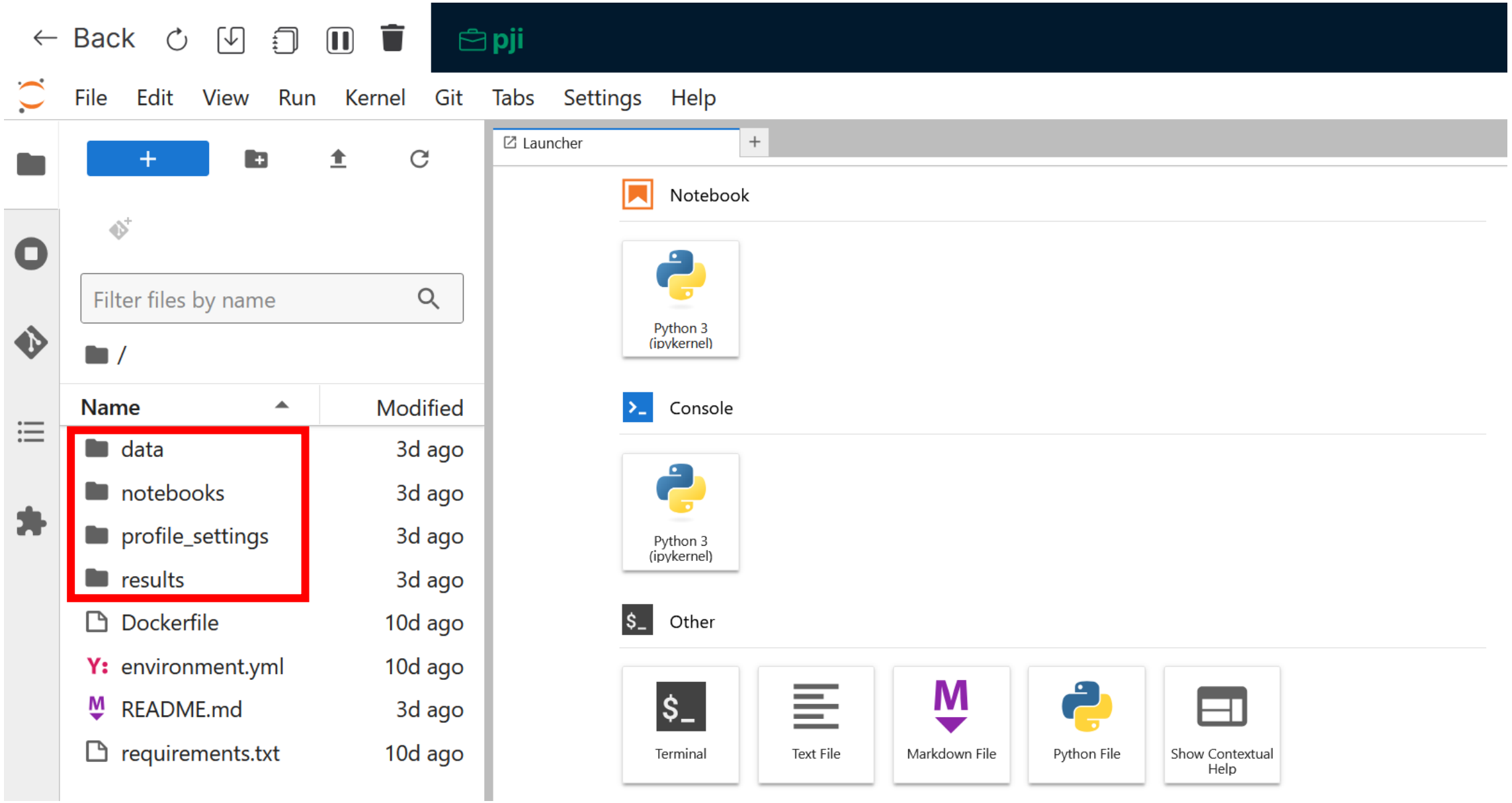
https://renkulab.io/projects/coline.mollaret/pji/sessions/new?autostart=1

It can take a few minutes to start a session.

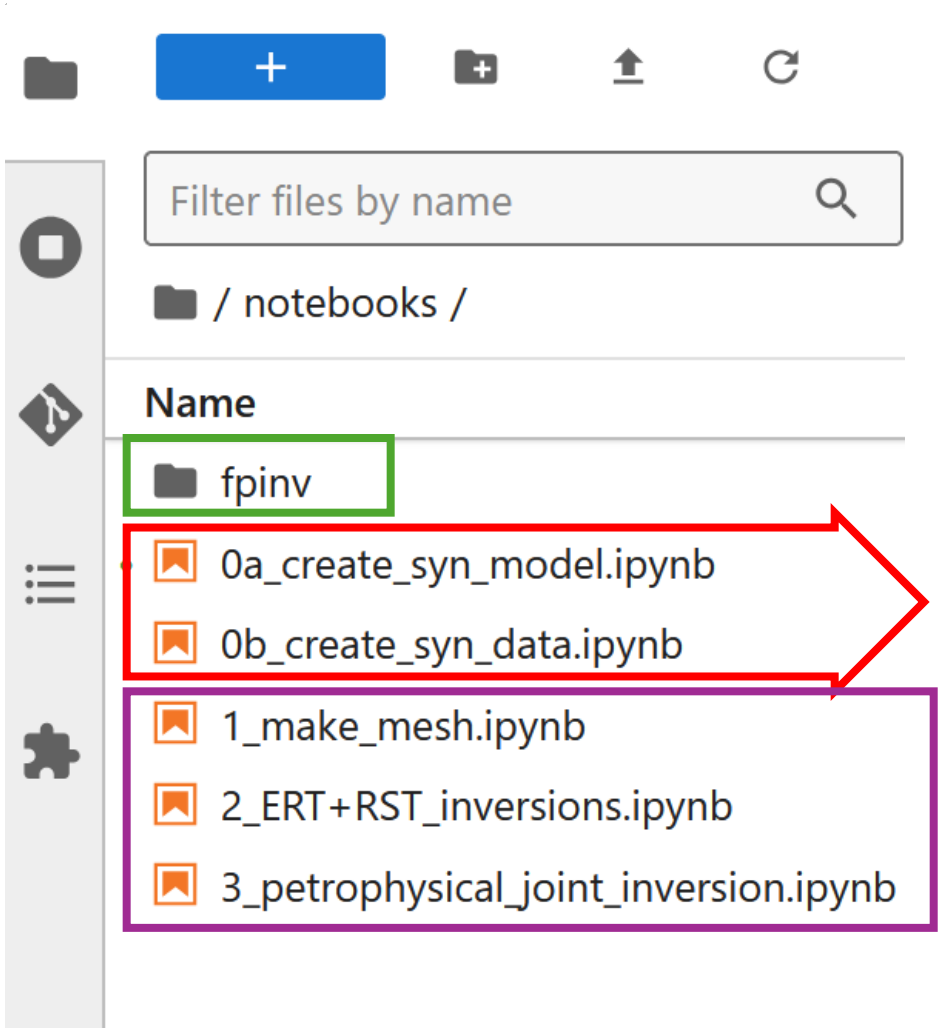




What do you find on the pji Renku project?



What do the notebook scripts do?

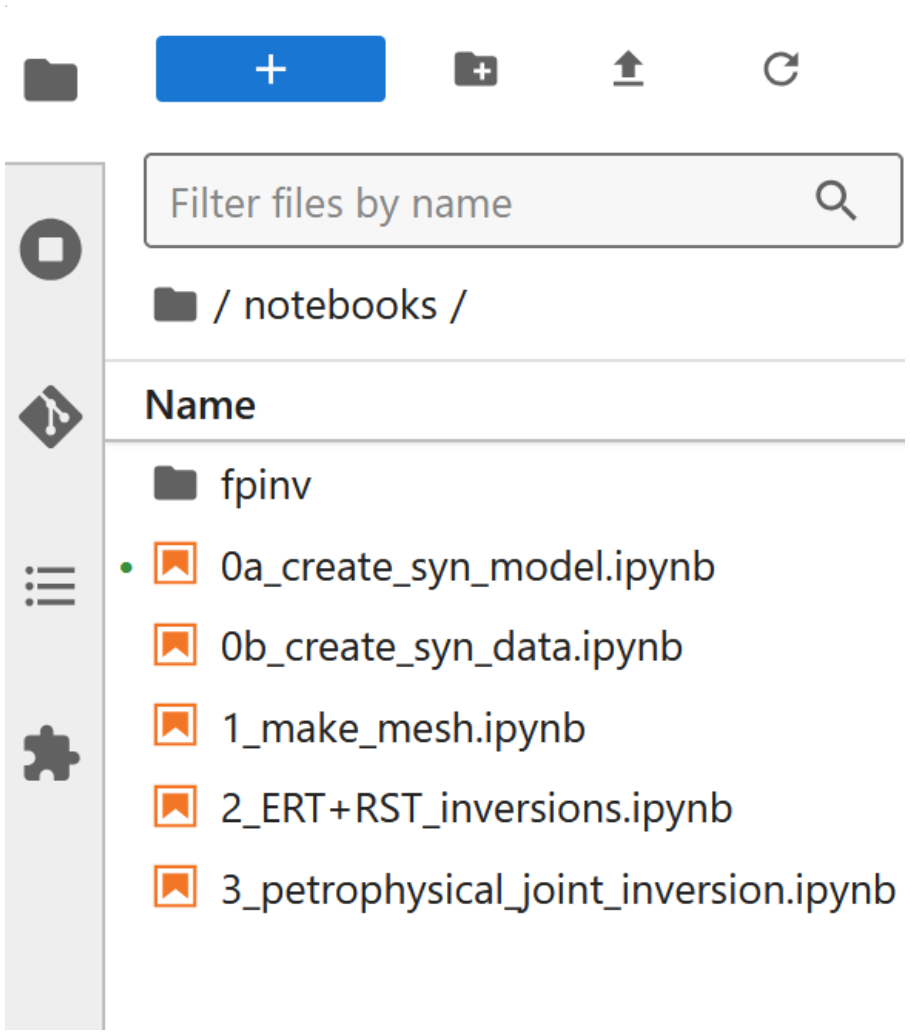


Folder containing core functions for the PJI

Only to create synthetic model and data

These 3 notebooks need to be run in the right order with the same dataset defined as «profile».

What to modify in the scripts?



- Choose your **profile name** (corresponding data are needed in the **data folder**!)
- Choose the **mesh or inversion parameters** mainly in the profile settings file (located in the **profile_settings** folder)

```
### PROFILE ##########################################
profile='CH_COR_V01_2007-07-13_01' # choose profile name
profile='CH_DOL_V01_2016-08-11_01'
profile='CH_LAP_MH1_2010-08-24_01'
#profile = 'KG_ABR_P03_2021-08-22_01'
#profile='3L'

print("profile: " + profile)
```

Mesh parametrization

1_make_mesh.ipynb

➢ **Play around with the mesh parametrization**

pygimli.meshtools.**createParaMesh2DGrid**(*sensors*, *paraDX=1*, *paraDZ=1*, *paraDepth=0*, *nLayers=11*, *boundary=-1*, *paraBoundary=2*, ***kwargs*) [source]

Create a grid-style mesh for an inversion parameter mesh.

Create a grid-style mesh for an inversion parameter mesh. Return parameter grid for a given list of sensor positions. Uses and forwards arguments to `pygimli.meshtools.appendTriangleBoundary`.

Parameters:

- **sensors** (*list of RVector3 objects or data container with sensorPositions*) – Sensor positions. Must be sorted in positive x direction
- **paraDX** (*float, optional*) – Horizontal distance between sensors, relative regarding sensor distance. Value must be greater than 0 otherwise 1 is assumed.
- **paraDZ** (*float, optional*) – Vertical distance to the first depth layer, relative regarding sensor distance. Value must be greater than 0 otherwise 1 is assumed.
- **paraDepth** (*float, optional*) – Maximum depth for parametric domain, 0 (default) means 0.4 * maximum sensor range.
- **nLayers** (*int, optional [11]*) – Number of depth layers.
- **boundary** (*int, optional [-1]*) – Boundary width to be appended for domain prolongation in absolute para domain width. Values lower than 0 force the boundary to be 4 times para domain width.
- **paraBoundary** (*int, optional [2]*) – Offset to the parameter domain boundary in absolute sensor spacing.

https://www.pygimli.org/

Individual inversions parametrization



2_ERT+RST_inversions.ipynb

Important parameters:

- Regularisation parameters
- Error estimates

- Try and compare different values of the **smoothness regularization parameter λ**
- Try and compare different values of **porosity** and the consequences in the applied 4PM estimates
- Try different data

COFFEE BREAK



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Petrophysical Joint Inversion parametrization

3_petrophysical_joint_inversion.ipynb

- Try and compare different values of the different material velocity and resistivity
- Try and compare different values of porosity (min, max, start)
- Try both the Archie (equa = 'A') and the Geometric mean model (equa = 'G')
- If wished, you can increase the max number of iterations
- Try out with your own data (?)

- Check the number of iterations
- Check the RMS error
- Check if the results make sense

And stay critical!

Play with synthetic data

- 0a_create_syn_model.ipynb
- 0b_create_syn_data.ipynb

➢Create your own synthetic model and then the corresponding data